VANI WALVEKAR

vaniw555@gmail.com | +1 (314) 3267687 |St. Louis, MO

linkedin.com/in/vani-walvekar-874938128 | GitHub: github.com/vani-walvekar1494 | Portfolio: vaniwalvekar.com/

PROFILE SUMMARY

Graduate student in Computer Science with research experience in **machine learning, high-performance computing, and scientific workflows**. Skilled in developing **reproducible, scalable ML pipelines** and **data-driven models** for large-scale simulations. Interested in **knowledge-guided AI and adaptive ML** for environmental and scientific applications, including **aquatic science, climate modeling, and ecosystem simulation**.

TECHNICAL SKILLS

- **Programming & Tools:** Python, SQL, JavaScript/TypeScript, React, Flask, Django, Docker, Git, CI/CD
- **HPC & Parallel Computing:** Nextflow, Apptainer/Singularity, Unix/Linux clusters, distributed workflows
- **Machine Learning:** TensorFlow, Scikit-learn, reinforcement learning, ML pipelines
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Cloud:** AWS Certified Cloud Practitioner, hybrid cloud integration

RESEARCH INTERESTS

- Machine learning
- Adaptive ML and data-efficient modeling
- HPC for accelerating scientific and environmental workflows
- Environmental and Earth system applications (aquatic science, wildfire, climate modeling)

CERTIFICATION, LEADERSHIP & AWARDS

- **AWS Certified Cloud Practitioner** – Aug 2025 (Valid until Aug 2028)
- **Teaching Assistant | Saint Louis University | June 2025 – Present**: Assisted in guiding and provided coding assistance to students in *Principles of Software Development* and *Databases*;
- **Winner**, UMSL Women's Hackathon – Built an innovative project with a cross-university team under a 24-hour deadline.
- **Full-Stack Developer**, SLU Launch Startup (MyB) - Developed a web app to monitor employee stress and cognitive well-being.

RESEARCH/ INDUSTRY EXPERIENCE

Software Developer | Open Source, Saint Louis University | Saint Louis, MO August 2025 - Present
eARM Scientific Workflow Enhancement

- Deployed and optimized molecular modeling workflows on Linux-based HPC clusters using Apptainer and Conda, improving reproducibility and scalability.
- Automated workflow execution and job scheduling with Nextflow, enhancing performance across distributed HPC environments.
- Troubleshot workflow errors, simulation convergence issues, and computational bottlenecks to ensure reliable results.
- Supported researchers and junior engineers in configuring, debugging, and running high-performance simulations.

Software Development Intern | Texas Advanced Computing Center | Austin, TX May 2025 – August 2025

- Developed HPC-backed **scientific dashboards** using React and Django, enhancing access to research data.
- Designed scalable backend **APIs and database models** for large-scale simulation data.
- Implemented **AI/ML-powered Storage Analyzer** to detect redundant data and recommend optimizations.
- Collaborated directly with scientists to **troubleshoot, integrate, and optimize HPC workflows**.

Associate Engineer | IBM(Kyndryl) | Bengaluru, India May 2019 – October 2022

- Integrated 200+ IBMi systems with AWS cloud, enabling hybrid cloud infrastructure deployment for global enterprise clients.
- Automated server maintenance using Python scripts, reducing manual workload by 30% and improving operational reliability.
- Resolved technical support tickets by troubleshooting hardware/software issues, SQL performance bottlenecks, and system errors.
- Ensured system compliance and security by managing OS upgrades, patching, and backup strategies.

Associate Software Engineer | Mphasis Ltd | Bengaluru, India December 2018 – April 2019

- Developed web applications using JavaScript, ASP.NET, Python (Flask/Django), enhancing functionality and engagement.
- Built responsive UIs with front-end frameworks, ensuring seamless cross-device user experiences.
- Collaborated with teams to deliver features and debug code, improving satisfaction and performance.

EDUCATION

M.S. in Computer Science | Saint Louis University | St. Louis, MO | GPA: 3.93 January 2024 – December 2025
B.E. in Computer Science | GSSSIETW | Mysuru, India | (First Class with Distinction) August 2014 - July 2018

ACADEMIC PROJECTS

Homeless Prevention Web Application (Flask | MySQL | Machine Learning Integration)
Built a Flask-based application integrating generative AI and machine learning models to predict and mitigate homelessness risks. Designed a responsive user interface, managed MySQL databases, and implemented CI/CD pipelines for optimized deployment.

Exoplanet Prediction (Python | Flask | Machine Learning | API Integration)
Built a reinforcement learning-based model to predict exoplanet characteristics from astronomical datasets, improving classification precision by 30%. Presented at the **GSA Symposium,** Saint Louis University.

Suicide Attempt Prediction (TensorFlow | Scikit-learn | data preprocessing | feature Engineering)
Built a machine learning model to predict suicide attempts, resulting in a 35% increase in prediction accuracy. I applied advanced data preprocessing and feature engineering techniques to enhance the model's performance.

Blind Hurdle Stick: IoT Navigation Aid (IoT | Computer Vision | Voice Control)
Developed a low-cost navigation aid for the visually impaired, integrating artificial vision and a voice-controlled panic system, improving user safety by 40% and autonomy by 35%. Presented the paper titled "Blind Hurdle Stick" at the **National Conference** at Christ University.

Segmentation of Text and Images from a Document (MATLAB | Image Processing | Pattern Recognition)
Built a page segmentation algorithm to separate textual and non-textual information from multimodal documents. Implemented noise removal preprocessing for better feature extraction and pattern recognition. Presented a paper titled "Segmentation of Text and Images from a Document" organized in association with **IEEE Bangalore**.

ESP32 LoRa Application Development (MicroPython | IoT | LoRa Communication)
Designed and implemented firmware for ESP32 LoRa devices, boosting IoT communication performance by 30%. Developed a real-time data transmission system using LoRa technology, enhancing monitoring capabilities by 20%.

PROFESSIONAL REFERENCES

Professional references available upon request